Linear Algebra — Lecture 6

Partitioned Matrices, Computing an Inverse, Solution Sets of Linear Systems

Friedberg–Insel–Spence, *Linear Algebra* 4e, §3.2–§3.4
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.2, §1.5, §2.2, §2.4

Topics. Partitioned matrices and block multiplication, the column–row expansion of AB, inverses of block matrices, the algorithm for computing A^{-1}, the solution sets of homogeneous and nonhomogeneous systems, and reading a solution set off the reduced row echelon form.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$, while Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Partitioned matrices	§2.4 (p. 117–118)	—
Column–row expansion	§2.4 (p. 119)	—
Inverse of a block matrix	§2.4 (p. 119–120)	—
Computing an inverse	§2.2 (p. 107–108)	§3.2 (p. 161–163)
Invertibility and rank	§2.3 (p. 111–112)	§3.2 (p. 152–159)
Homogeneous systems	§1.5 (p. 43–47)	§3.3 (p. 171–172)
Nonhomogeneous systems	§1.5 (p. 43–47)	§3.3 (p. 172–174)
Reduced echelon form, basic variables	§1.2 (p. 12–21)	§3.4 (p. 185–190)

Two questions that drive the whole lecture: (1) *After a matrix is cut into blocks, do multiplication and inversion still work block by block?* (2) *Can inverting a matrix and solving a system share one row reduction, and which step is thereby saved?*

How to read: use Lay first to get the algorithm under your fingers, then Friedberg to see why each step is legitimate.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 Partitioned Matrices

The first five lectures followed one line: vector spaces, linear transformations, matrix representations. Every linear map became a matrix, a change of basis became Q on one side and Q^{-1} on the other, and the rank was the one number that survived the whole line. This lecture changes the scale: it looks at the **blocks** of a matrix rather than at its entries. Once a large matrix is cut into blocks, addition, scalar multiplication and multiplication all proceed block by block, and inversion reduces to two inverses of smaller matrices and two multiplications. The LU factorization of Lecture 7 is the same idea applied to row reduction: collect the multipliers used during elimination into one lower triangular matrix, and what comes out is a decomposition that can be reused.

Definition 1.1 (Partitioned matrix; Lay §2.4, p. 117)**.** Let $A \in M_{m\times n}(\mathbb{F})$. Draw several horizontal and vertical lines through A; each piece so obtained is a **submatrix** of A, the arrangement is a **partition** of A, and A itself is called a **partitioned matrix**. If the rows are cut into p groups and the columns into q groups, write $A = [A_{ij}]$, where A_{ij} is the submatrix lying at the i-th row group and the j-th column group.

Addition and **scalar multiplication** of partitioned matrices are performed block by block: provided the two matrices have the same size and are partitioned in exactly the same way, every block of $A + B$ is the sum of the corresponding blocks, and every block of cA is cA_{ij}.

Definition 1.2 (Conformable partitions; Lay §2.4, p. 118)**.** Let A be $m \times n$ and B be $n \times k$. If the **column** partition of A and the **row** partition of B break at the same points, the two partitions are **conformable for block multiplication**.

Theorem 1.1 (Block multiplication; Lay §2.4, p. 118)**.** *Let A be $m \times n$ and B be $n \times k$ with conformable partitions: the rows of A are cut into $m_1, \dots, m_p$ groups, the columns of A and the rows of B are both cut into $n_1, \dots, n_q$ groups, and the columns of B are cut into $k_1, \dots, k_r$ groups. Then AB, partitioned in the same way, is $[\,(AB)_{ij}\,]$ with*

$$(AB)_{ij} = \sum_{l=1}^{q} A_{il} B_{lj} \qquad (i = 1, \dots, p;\ j = 1, \dots, r).$$

Proof. Compare entries. The (x, y)-entry of AB is $\sum_{t=1}^{n} a_{xt} b_{ty}$. Group the index t by column groups: the l-th group pairs row x of A_{il} with column y of B_{lj} and adds the products, which is exactly the (x, y)-entry of $A_{il} B_{lj}$. Summing over the q groups gives the (x, y)-entry of $\sum_l A_{il} B_{lj}$, term by term equal to $\sum_t a_{xt} b_{ty}$. □

Theorem 1.2 (Column–row expansion; Lay Theorem 10, §2.4, p. 119)**.** *Let A be $m \times n$ and B be $n \times k$, and let $\mathrm{col}_k(A)$ be the k-th column of A ($m \times 1$) and $\mathrm{row}_k(B)$ the k-th row of B ($1 \times k$). Then*

$$AB = \sum_{k=1}^{n} \mathrm{col}_k(A)\ \mathrm{row}_k(B) = \begin{bmatrix} \mathrm{col}_1(A) & \cdots & \mathrm{col}_n(A) \end{bmatrix} \begin{bmatrix} \mathrm{row}_1(B) \\ \vdots \\ \mathrm{row}_n(B) \end{bmatrix}$$

Proof. The right-hand side is the block product for the partition of A into single columns and the partition of B into single rows, and those partitions are conformable. By Theorem 1.1 its (i, j)-entry is $\sum_k a_{ik} b_{kj}$, which is the (i, j)-entry of AB. □

Each block $\mathrm{col}_k(A)\,\mathrm{row}_k(B)$ in the brackets is an $m \times k$ matrix, a product of one column and one row (an **outer product**); the theorem writes AB as a sum of n matrices of rank at most 1. This is the fifth reading of AB; the other four are $A\boldsymbol{x}$ computed from the columns of A, the columns of AB computed from the columns of B, the entry-by-entry row-column rule, and the rows of AB computed from the rows of A; all five are catalogued in Lay §2.4, p. 118.

Definition 1.3 (Block upper triangular and block diagonal; Lay §2.4, p. 119–120)**.** Let A be a partitioned matrix. If every block below the main diagonal is a zero matrix, that is, if A has the form

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix},$$

then A is a **block upper triangular matrix**; if every block off the main diagonal is a zero matrix, then A is a **block diagonal matrix**.

Proposition 1.3 (Inverses of block triangular matrices; Lay Example 5, §2.4, p. 119–120)**.** *Let*

$$A = \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix}$$

be block upper triangular, with A_{11} of size $p \times p$ and A_{22} of size $q \times q$. Then A is invertible if and only if both A_{11} and A_{22} are invertible, and in that case

$$A^{-1} = \begin{bmatrix} A_{11}^{-1} & -A_{11}^{-1} A_{12} A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{bmatrix}$$

In particular a block diagonal matrix is invertible if and only if every diagonal block is invertible, and its inverse is the block diagonal matrix of the inverses.

Proof. Suppose first that A_{11} and A_{22} are invertible and call the matrix on the right B. Compute AB and BA block by block: the upper-left block is $A_{11}A_{11}^{-1} = I_p$, the lower-right block is $A_{22}A_{22}^{-1} = I_q$, the lower-left block is $0 \cdot A_{11}^{-1} + A_{22} \cdot 0 = 0$, and the upper-right block is $A_{11}\big(-A_{11}^{-1}A_{12}A_{22}^{-1}\big) + A_{12}A_{22}^{-1} = -A_{12}A_{22}^{-1} + A_{12}A_{22}^{-1} = 0$; the four blocks of BA are the same. Hence $B = A^{-1}$.

Suppose now that A is invertible. Partition A^{-1} in the same way into four blocks B_{ij} (with B_{11} of size $p \times p$); from $AA^{-1} = I_{p+q}$ come the four block equations

$$A_{11}B_{11} + A_{12}B_{21} = I_p, \quad A_{11}B_{12} + A_{12}B_{22} = 0, \quad A_{22}B_{21} = 0, \quad A_{22}B_{22} = I_q.$$

The last one, together with "a square matrix with a one-sided inverse is invertible" (Lecture 4 notes, Theorem 3.7), makes A_{22} invertible with $B_{22} = A_{22}^{-1}$. The third then gives $B_{21} = A_{22}^{-1} \cdot 0 = 0$, so the first becomes $A_{11}B_{11} = I_p$, and the same argument makes A_{11} invertible with $B_{11} = A_{11}^{-1}$. Finally the second equation reads $A_{11}B_{12} = -A_{12}A_{22}^{-1}$, and left multiplication by A_{11}^{-1} gives $B_{12} = -A_{11}^{-1}A_{12}A_{22}^{-1}$. □

Example 1.1 (Block multiplication and a block inverse; written for these notes; method from Lay §2.4, p. 117–120)**.** Let

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

partitioned into 2×2 blocks (the first two rows and columns, then the last two). Find A^{-1} block by block, and check the answer by multiplying the two 4×4 matrices with the same partition.

Solution. With the $2 + 2$ partition,

$$A_{11} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad A_{12} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2, \quad A_{21} = 0, \quad A_{22} = \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}$$

The two diagonal blocks have determinants $2 \cdot 1 - 1 \cdot 1 = 1$ and $3 \cdot 1 - 1 \cdot 1 = 2$, both nonzero, so

$$A_{11}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}, \qquad A_{22}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix}$$

The upper-right block is $-A_{11}^{-1}A_{12}A_{22}^{-1} = -A_{11}^{-1}A_{22}^{-1}$; compute the product first,

$$A_{11}^{-1}A_{22}^{-1} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ -1 & 3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 2 & -4 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -\frac{3}{2} & \frac{7}{2} \end{bmatrix}$$

and change its sign:

$$A^{-1} = \left[\begin{array}{cc|cc} 1 & -1 & -1 & 2 \\ -1 & 2 & \frac{3}{2} & -\frac{7}{2} \\ \hline 0 & 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & -\frac{1}{2} & \frac{3}{2} \end{array}\right]$$

All four are right, so $AA^{-1} = I_4$. □

Approach. The lower-left block is zero, so A is block upper triangular and Proposition 1.3 writes the inverse down: two 2×2 inverses and two 2×2 products suffice. Seeing a whole block of zeros, partition first; a 4×4 problem then turns into three 2×2 problems, which is half the work of row reducing this matrix directly.

2 Computing the Inverse

Lecture 4 established that an $n \times n$ matrix is invertible if and only if it is row equivalent to I_n. That theorem answers yes or no; this section opens its proof to read off an algorithm: attach I_n to the right of A, row reduce the whole array, and when the left half has become I_n the right half has become A^{-1}.

Definition 2.1 (Augmented matrix; Friedberg §3.2, p. 161)**.** Let A be $m \times n$ and B be $m \times p$. The $m \times (n+p)$ matrix obtained by placing B to the right of A is written $(A \,|\, B)$ and called the **augmented matrix** of A and B: its first n columns are the columns of A, its last p columns the columns of B.

Theorem 2.1 (An inverse by row reduction; Friedberg §3.2, p. 161–162)**.** *Let A be an $n \times n$ matrix.*

(1) If A is invertible, then elementary row operations carry $(A \,|\, I_n)$ to $(I_n \,|\, A^{-1})$.

(2) Conversely, if some sequence of elementary row operations carries $(A \,|\, I_n)$ to $(I_n \,|\, B)$, then A is invertible and $B = A^{-1}$.

Proof. (1) By Lecture 5 notes, Proposition 3.8, an invertible matrix is a product of elementary matrices; write $A^{-1} = E_p \cdots E_1$. Left multiplication by an elementary matrix is an elementary row operation, so

$$E_p \cdots E_1 (A \,|\, I_n) = (A^{-1}A \,|\, A^{-1}I_n) = (I_n \,|\, A^{-1}),$$

that is, the sequence carries $(A \,|\, I_n)$ to $(I_n \,|\, A^{-1})$.

(2) Let $E_1, \ldots, E_p$ be the elementary matrices of that sequence and put $M = E_p \cdots E_1$. Then $M(A \,|\, I_n) = (MA \,|\, M) = (I_n \,|\, B)$, so comparing the two halves gives $MA = I_n$ and $M = B$. By "a square matrix with a one-sided inverse is invertible" (Lecture 4 notes, Theorem 3.7), A is invertible with $M = A^{-1}$, hence $B = A^{-1}$. □

Remark 2.1 (What failure looks like; Friedberg §3.2, p. 162)**.** If A is not invertible, every attempt to carry $(A \,|\, I_n)$ to $(I_n \,|\, B)$ must fail: elementary row operations preserve rank (Lecture 5 notes, Theorem 3.4), and $\operatorname{rank}(A) < n$, so the left half produces at most $n-1$ pivots and some row whose first n entries are all zero must appear. Once such a row shows up there is no point continuing: A is not invertible.

Example 2.1 (An inverse by row reduction; Friedberg Example 5, §3.2, p. 162–163)**.** Decide whether

$$A = \begin{bmatrix} 0 & 2 & 4 \\ 2 & 4 & 2 \\ 3 & 3 & 1 \end{bmatrix}$$

is invertible, and if it is, find A^{-1}.

Solution.

$$\left[\begin{array}{ccc|ccc} 0 & 2 & 4 & 1 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 2 & 4 & 2 & 0 & 1 & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & \frac{1}{2} & 0 \\ 0 & 2 & 4 & 1 & 0 & 0 \\ 0 & -3 & -2 & 0 & -\frac{3}{2} & 1 \end{array}\right]$$

(interchange rows 1 and 2; multiply row 1 by $\frac{1}{2}$; subtract three times row 1 from row 3). Then

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{1}{2} & 0 & 0 \\ 0 & -3 & -2 & 0 & -\frac{3}{2} & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -1 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 4 & \frac{3}{2} & -\frac{3}{2} & 1 \end{array}\right]$$

(multiply row 2 by $\frac{1}{2}$; subtract twice row 2 from row 1; add three times row 2 to row 3). Finally

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -1 & \frac{1}{2} & 0 \\ 0 & 1 & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{8} & -\frac{5}{8} & \frac{3}{4} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{2} \\ 0 & 0 & 1 & \frac{3}{8} & -\frac{3}{8} & \frac{1}{4} \end{array}\right]$$

(multiply row 3 by $\frac{1}{4}$; add three times row 3 to row 1; subtract twice row 3 from row 2). The left half is I_3, so A is invertible and

$$A^{-1} = \frac{1}{8}\begin{bmatrix} 1 & -5 & 6 \\ -2 & 6 & -4 \\ 3 & -3 & 2 \end{bmatrix}$$ □

Approach. Attach I_3 on the right and row reduce only the left half, letting the right half move along. The first entry is 0, so interchange rows 1 and 2; then clear each column in turn, scaling the pivot to 1 and eliminating the rest of its column. When the left half has become I_3, the right half is A^{-1}; if a whole zero row appears on the left, A is not invertible.

3 Solution Sets of Linear Systems

The previous section answered "when does $A\boldsymbol{x} = \boldsymbol{b}$ have exactly one solution", and it used the inverse of an $n \times n$ matrix to do it. Widening the same question to an arbitrary $m \times n$ A starts with the shape of the solution set itself. The result of this section fits in one sentence: **the solution set of a homogeneous system is the kernel of L_A, and the solution set of a nonhomogeneous system is that kernel shifted by one particular solution**. Lay describes the same thing in §1.5 with free variables; here that description is rewritten in the language of subspaces.

Definition 3.1 (Homogeneous and nonhomogeneous systems; Friedberg §3.3, p. 171)**.** Let $A \in M_{m\times n}(\mathbb{F})$ and $\boldsymbol{b} \in \mathbb{F}^m$. The system $A\boldsymbol{x} = \boldsymbol{b}$ is **homogeneous** when $\boldsymbol{b} = \boldsymbol{0}$ and **nonhomogeneous** when $\boldsymbol{b} \neq \boldsymbol{0}$. The matrix A defines the left-multiplication map $L_A : \mathbb{F}^n \to \mathbb{F}^m$, $L_A(\boldsymbol{x}) = A\boldsymbol{x}$, so that

$$A\boldsymbol{x} = \boldsymbol{b} \iff L_A(\boldsymbol{x}) = \boldsymbol{b}, \qquad A\boldsymbol{x} = \boldsymbol{0} \iff L_A(\boldsymbol{x}) = \boldsymbol{0}.$$

A homogeneous system always has the solution $\boldsymbol{x} = \boldsymbol{0}$. Write K for the solution set of $A\boldsymbol{x} = \boldsymbol{b}$ and K_H for the solution set of the corresponding homogeneous system.

Theorem 3.1 (A homogeneous solution set is the kernel; Friedberg Theorem 3.8, §3.3, p. 171)**.** *Let $A\boldsymbol{x} = \boldsymbol{0}$ be a homogeneous system of m equations in n unknowns and let K be its solution set. Then*

$$K = \mathcal{N}(L_A),$$

so K is a subspace of $\mathbb{F}^n$, and

$$\dim K = n - \operatorname{rank}(A).$$

Proof. $\boldsymbol{x} \in K$ if and only if $A\boldsymbol{x} = \boldsymbol{0}$, that is $L_A(\boldsymbol{x}) = \boldsymbol{0}$, that is $\boldsymbol{x} \in \mathcal{N}(L_A)$; the kernel of a matrix transformation is the null space of A (Lecture 3 notes, Proposition 2.2), and a kernel is a subspace (Lecture 3 notes, Theorem 2.1). The dimension follows from the Dimension Theorem (Lecture 3 notes, Theorem 2.4) together with $\operatorname{rank}(A) = \dim \mathcal{R}(L_A)$. □

Proposition 3.2 (More unknowns than equations forces a nonzero solution; Friedberg §3.3, p. 171, Corollary)**.** *Let A be an $m \times n$ matrix. If $m < n$, then $A\boldsymbol{x} = \boldsymbol{0}$ has a nonzero solution.*

Proof. $\operatorname{rank}(A) \leq m$, so by Theorem 3.1, $\dim K = n - \operatorname{rank}(A) \geq n - m > 0$, hence $K \neq \{\boldsymbol{0}\}$. □

Theorem 3.3 (A nonhomogeneous solution set is a shift of the kernel; Friedberg Theorem 3.9, §3.3, p. 172)**.** *Let K be the solution set of $A\boldsymbol{x} = \boldsymbol{b}$, let K_H be the solution set of the corresponding system $A\boldsymbol{x} = \boldsymbol{0}$, and let $\boldsymbol{s}$ be any solution of $A\boldsymbol{x} = \boldsymbol{b}$. Then*

$$K = \{\boldsymbol{s}\} + K_H = \{\boldsymbol{s} + \boldsymbol{k} : \boldsymbol{k} \in K_H\}.$$

Proof. First $K \subseteq \{\boldsymbol{s}\} + K_H$. Take $\boldsymbol{w} \in K$; then $A\boldsymbol{w} = \boldsymbol{b} = A\boldsymbol{s}$, so $A(\boldsymbol{w} - \boldsymbol{s}) = \boldsymbol{0}$, that is $\boldsymbol{w} - \boldsymbol{s} \in K_H$, and $\boldsymbol{w} = \boldsymbol{s} + \boldsymbol{k}$ with $\boldsymbol{k} = \boldsymbol{w} - \boldsymbol{s}$. Conversely, if $\boldsymbol{w} = \boldsymbol{s} + \boldsymbol{k}$ with $\boldsymbol{k} \in K_H$, then $A\boldsymbol{w} = A\boldsymbol{s} + A\boldsymbol{k} = \boldsymbol{b} + \boldsymbol{0} = \boldsymbol{b}$, so $\boldsymbol{w} \in K$. □

Remark 3.1 (How to use the two sentences about solution sets)**.** Theorem 3.1 says the homogeneous solution set is a subspace **through the origin**; Theorem 3.3 says the nonhomogeneous solution set is that subspace **shifted** onto a particular solution. How many solutions a nonhomogeneous system has is therefore decided entirely by K_H: $K_H = \{\boldsymbol{0}\}$ means a unique solution, and a nonzero vector in K_H means as many solutions as $\mathbb{F}$ has elements.

For an $n \times n$ matrix, $K_H = \{\boldsymbol{0}\}$ is exactly invertibility; the solution set is then the single point $\boldsymbol{x} = A^{-1}\boldsymbol{b}$, which is Lecture 4 notes, Theorem 3.2. The converse holds too: if $A\boldsymbol{x} = \boldsymbol{b}$ has exactly one solution for every $\boldsymbol{b}$, then A is invertible (Friedberg Theorem 3.10, §3.3, p. 174).

Example 3.1 (Solving a system with the inverse of the previous section; Friedberg Example 4, §3.3, p. 174)**.** Solve the system

$$2x_2 + 4x_3 = 2, \qquad 2x_1 + 4x_2 + 2x_3 = 3, \qquad 3x_1 + 3x_2 + x_3 = 1;$$

its coefficient matrix is the A whose inverse was computed in Example 2.1.

Solution. Write the system as $A\boldsymbol{x} = \boldsymbol{b}$ with $\boldsymbol{b} = (2, 3, 1)^\top$, and recall

$$A^{-1} = \frac{1}{8}\begin{bmatrix} 1 & -5 & 6 \\ -2 & 6 & -4 \\ 3 & -3 & 2 \end{bmatrix}.$$

Then

$$\boldsymbol{x} = A^{-1}\boldsymbol{b} = \frac{1}{8}\begin{bmatrix} 1\cdot 2 - 5\cdot 3 + 6\cdot 1 \\ -2\cdot 2 + 6\cdot 3 - 4\cdot 1 \\ 3\cdot 2 - 3\cdot 3 + 2\cdot 1 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -7 \\ 10 \\ -1 \end{bmatrix} = \left(-\tfrac{7}{8},\ \tfrac{5}{4},\ -\tfrac{1}{8}\right)^\top.$$ □

Approach. Look at the coefficient matrix first. Its determinant is nonzero and the previous section already produced A^{-1}, so by Lecture 4 notes, Theorem 3.2 the system has the unique solution $\boldsymbol{x} = A^{-1}\boldsymbol{b}$: multiply the inverse by the right-hand side and no second row reduction is needed. This is the case $K_H = \{\boldsymbol{0}\}$ of Theorem 3.3.

The theorems above write the solution set as $\{\boldsymbol{s}\} + K_H$ but do not say where $\boldsymbol{s}$ and a basis of K_H come from. What follows supplies the procedure: reduce the augmented matrix to **reduced row echelon form**; the pivot columns give the basic variables and the remaining columns the free variables; setting every free variable to 0 gives a particular solution, and setting one of them to 1 at a time gives a basis of K_H. The number of vectors obtained is $n - r$, exactly the dimension the Dimension Theorem predicts.

Definition 3.2 (Reduced row echelon form; Friedberg §3.4, p. 185)**.** A matrix is in **reduced row echelon form** if all three of the following hold:

(a) every row containing a nonzero entry precedes every row that is entirely zero;
(b) the first nonzero entry of each row is the only nonzero entry of its column;
(c) the first nonzero entry of each row equals 1 and lies in a column to the right of the first nonzero entry of the preceding row.

Theorem 3.4 (Invertible left multiplication preserves the solution set; Friedberg Theorem 3.13, §3.4, p. 182)**.** *Let $A\boldsymbol{x} = \boldsymbol{b}$ be a system of m equations in n unknowns and let C be an invertible $m \times m$ matrix. Then $CA\boldsymbol{x} = C\boldsymbol{b}$ and $A\boldsymbol{x} = \boldsymbol{b}$ have the same solutions.*

Proof. Let K and K' be the two solution sets. If $\boldsymbol{w} \in K$, then $A\boldsymbol{w} = \boldsymbol{b}$, and left multiplication by C gives $CA\boldsymbol{w} = C\boldsymbol{b}$, so $\boldsymbol{w} \in K'$. Conversely, if $\boldsymbol{w} \in K'$, then $CA\boldsymbol{w} = C\boldsymbol{b}$; left multiplication by C^{-1} gives $A\boldsymbol{w} = C^{-1}C\boldsymbol{b} = \boldsymbol{b}$, so $\boldsymbol{w} \in K$. □

Proposition 3.5 (Elementary row operations preserve the solution set; Friedberg §3.4, p. 182, Corollary)**.** *If $(\,A' \mid \boldsymbol{b}'\,)$ is obtained from $(\,A \mid \boldsymbol{b}\,)$ by finitely many elementary row operations, then $A'\boldsymbol{x} = \boldsymbol{b}'$ and $A\boldsymbol{x} = \boldsymbol{b}$ have the same solutions.*

Proof. Each elementary row operation is left multiplication by an elementary matrix (Lecture 5 notes, Theorem 3.1), and elementary matrices are invertible (Lecture 5 notes, Theorem 3.2), so each step leaves the solution set unchanged by Theorem 3.4; finitely many such steps leave it unchanged as well. □

Algorithm 3.1 (Reading a solution set off the reduced echelon form)**.**

Input: the augmented matrix $(\,A \mid \boldsymbol{b}\,)$, already in reduced row echelon form.
Output: the solution set of $A\boldsymbol{x} = \boldsymbol{b}$.

1. *Record the pivot columns $j_1 < \cdots < j_r$ and the free columns $f_1 < \cdots < f_{n-r}$; write the i-th nonzero row as $x_{j_i} + c_{i1}x_{f_1} + \cdots + c_{i,n-r}x_{f_{n-r}} = \beta_i$, where β_i is its entry in the augmented column;*
2. *set the free variables equal to parameters, $x_{f_k} = t_k$, and read each basic variable off its own row: $x_{j_i} = \beta_i - c_{i1}t_1 - \cdots - c_{i,n-r}t_{n-r}$;*
3. *with all $t_k = 0$, obtain the particular solution $\boldsymbol{s}_0$: β_i in the pivot positions and 0 in the free ones;*
4. *for each $k = 1, \ldots, n-r$, take $t_k = 1$ and the other parameters 0 to obtain the vector $\boldsymbol{u}_k$: its f_k-th component is 1, its other free components are 0, and its j_i-th component is $-c_{ik}$ (where c_{ik} is the entry of the i-th nonzero row in column f_k);*
5. *output the solution set $\boldsymbol{s}_0 + \operatorname{Span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-r}\}$, where $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-r}\}$ is a basis of K_H.*

Theorem 3.6 (Reading a solution set off the reduced echelon form; Friedberg Theorem 3.15, §3.4, p. 189–190)**.** *Let $A\boldsymbol{x} = \boldsymbol{b}$ be a system of r nonzero equations in n unknowns with $\operatorname{rank}(A) = \operatorname{rank}(A \mid \boldsymbol{b})$, and suppose $(\,A \mid \boldsymbol{b}\,)$ is in reduced row echelon form. Then*

(a) $\operatorname{rank}(A) = r$*;*
(b) if the general solution produced by Algorithm 3.1 is written as $\boldsymbol{s} = \boldsymbol{s}_0 + t_1\boldsymbol{u}_1 + \cdots + t_{n-r}\boldsymbol{u}_{n-r}$, then $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-r}\}$ is a basis of K_H and $\boldsymbol{s}_0$ is a solution of the original system.

Proof. (a) A matrix in reduced row echelon form has exactly r nonzero rows, and by definition those rows are linearly independent, so $\operatorname{rank}(A \mid \boldsymbol{b}) = r$, hence $\operatorname{rank}(A) = r$.

(b) Setting $t_1 = \cdots = t_{n-r} = 0$ gives $\boldsymbol{s} = \boldsymbol{s}_0$, so $\boldsymbol{s}_0$ solves the original system. By Theorem 3.3, $K = \{\boldsymbol{s}_0\} + K_H$, hence $K_H = \{-\boldsymbol{s}_0\} + K = \operatorname{Span}\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-r}\}$. Also $\dim K_H = n - \operatorname{rank}(A) = n - r$ (Theorem 3.1), and the spanning set has only $n - r$ vectors, so it is a basis of K_H. □

Example 3.2 (Reading a general solution off the reduced echelon form; classroom example of 30 October 2026)**.** The augmented matrix of a system has already been reduced to

$$\left[\begin{array}{ccccc|c} 1 & 0 & -2 & 3 & 0 & -24 \\ 0 & 1 & -2 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Write down its general solution and a basis of the solution set K_H of the corresponding homogeneous system.

Solution. The pivots lie in columns $1, 2, 5$, so x_1, x_2, x_5 are the **basic variables** and x_3, x_4 the **free variables**; there are 3 nonzero rows, so $r = 3$ and there are $n - r = 2$ free variables.

Solve each basic variable from the equation of its own row:

$$x_1 = -24 + 2x_3 - 3x_4, \qquad x_2 = -7 + 2x_3 - 2x_4, \qquad x_5 = 4.$$

Treating x_3, x_4 as parameters t_1, t_2, the general solution is

$$\boldsymbol{s} = \begin{bmatrix} -24 \\ -7 \\ 0 \\ 0 \\ 4 \end{bmatrix} + t_1 \begin{bmatrix} 2 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} -3 \\ -2 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \boldsymbol{s}_0 + t_1 \boldsymbol{u}_1 + t_2 \boldsymbol{u}_2,$$

where $\boldsymbol{s}_0$ is the particular solution at $t_1 = t_2 = 0$ and $\boldsymbol{u}_1, \boldsymbol{u}_2$ solve the homogeneous system. □

Approach. Faced with a reduced row echelon form, count the pivots to name the basic variables; the remaining columns are the free ones. Treat the free variables as parameters and solve each row for its own basic variable. The particular solution sets every parameter to zero; the basis vectors set one parameter to 1 and the rest to zero. The $n - r$ vectors obtained this way are automatically independent, because their free-variable components are the standard basis vectors, and $\dim K_H = n - r$, so there is nothing left to check about spanning.

Remark 3.2 (The solution set of the matrix equation $AX = B$)**.** Let $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{m\times p}(\mathbb{F})$, the unknown being $X \in M_{n\times p}(\mathbb{F})$. Write the columns of B as $\boldsymbol{b}_1, \dots, \boldsymbol{b}_p$; then $AX = B$ amounts to the p systems $A\boldsymbol{x}_j = \boldsymbol{b}_j$ holding at once, the j-th column of X being the solution of the j-th system.

So a single pass with $[\,A \mid B\,]$ to reduced echelon form suffices; let the result be $[\,R \mid X_0\,]$, where R is the reduced echelon form of A and the columns of X_0 are particular solutions of the p systems. When no column of B is a pivot column the systems are consistent, equivalently $\operatorname{rank}(A) = \operatorname{rank}([\,A \mid B\,])$. The solution set is then

$$\{\, X_0 + NC : C \in M_{(n-r)\times p}(\mathbb{F}) \,\},$$

where C is any matrix in $M_{(n-r)\times p}(\mathbb{F})$, the columns of N form a basis of $K_H = \mathcal{N}(L_A)$, and $r = \operatorname{rank}(A)$.

For $p = 1$ this reduces to the vector equation and agrees with the general solution $\boldsymbol{s}_0 + t_1\boldsymbol{u}_1 + \cdots$ of Theorem 3.6, where $\boldsymbol{u}_1, \dots, \boldsymbol{u}_{n-r}$ is a basis of $K_H = \mathcal{N}(L_A)$. When A is invertible the solution set has the single element $A^{-1}B$.

4 A Worked Example

Example 4.1 (Failing to invert, and the shape of solution sets; written for these notes; methods from Friedberg §3.2, p. 161–163 and §3.3–§3.4)**.** Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 1 & 2 & 2 \end{bmatrix}, \qquad \boldsymbol{b} = (1, 2, 1)^{\top}, \qquad \boldsymbol{b}' = (1, 2, 0)^{\top}.$$

(1) Use $[\,A \mid I_3\,]$ to find A^{-1}, and say where the reduction gets stuck if it cannot be found; (2) solve $A\boldsymbol{x} = \boldsymbol{b}$, writing the solution set and a basis of K_H; (3) solve $A\boldsymbol{x} = \boldsymbol{b}'$.

Solution. **(1) Inverting.** Put A next to I_3 and reduce:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 4 & 7 & 0 & 1 & 0 \\ 1 & 2 & 2 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 0 & -\frac{2}{3} & \frac{7}{3} \\ 0 & 0 & 1 & 0 & \frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 0 & 1 & -\frac{1}{3} & -\frac{1}{3} \end{array}\right].$$

A zero row appears on the left, so A has rank 2, is not invertible, and A^{-1} does not exist.

(2) Solving $A\boldsymbol{x} = \boldsymbol{b}$. Replace the right-hand side by $\boldsymbol{b}$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & 7 & 2 \\ 1 & 2 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

so $x_1 = 1 - 2x_2$ and $x_3 = 0$, with x_2 free. Taking $x_2 = 0$ gives the particular solution $\boldsymbol{s}_0 = (1, 0, 0)^\top$, and a basis of the homogeneous solution set is $\boldsymbol{u} = (-2, 1, 0)^\top$; hence

$$\{\boldsymbol{x}\} = \boldsymbol{s}_0 + \mathrm{Span}\{\boldsymbol{u}\} = \big\{(1,0,0)^\top + t(-2,1,0)^\top : \ t \in \mathbb{F}\big\},$$

with $K_H = \mathrm{Span}\{(-2,1,0)^\top\}$ of dimension 1.

(3) Solving $A\boldsymbol{x} = \boldsymbol{b}'$. Replace the right-hand side by $\boldsymbol{b}'$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 4 & 7 & 2 \\ 1 & 2 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right],$$

whose last row reads $0 = 1$: the system is inconsistent and has no solution. □

Approach. One matrix, two right-hand sides, three outcomes. When A is not invertible the left block ends with a zero row; when the right-hand side lies in the column space the system is consistent and the solution set is a particular solution plus the kernel, with as many free parameters as free variables; when it does not, a row $0 = 1$ appears.

5 Supplement: notions Homework 6 uses but the lecture did not cover

Remark 5.1 (Where these come from)**.** None of the following sits on the lecture's main line, yet Homework 6 uses it:

- the **annihilator** S^0: Homework sheet, Definition 1; Friedberg §2.6, the preamble to Exercises 13–17 (printed page 126);
- the **canonical map** ψ: the second half of Definition 1 on the sheet. It is the ψ of Lecture 5 notes, Theorem 2.4, that is $\psi(\boldsymbol{x})(f) = f(\boldsymbol{x})$;
- **the transpose of a transpose** $(T^\top)^\top$: Homework sheet, Problem 1(f); apply the transpose of Lecture 5 notes, Theorem 2.2 twice;
- **the sum of subsets** $W_1 + W_2$: Homework sheet, Problem 11(a); the definition is in Friedberg §1.3 exercises (printed page 22).

Definition 5.1 (Annihilator; Homework sheet, Definition 1; Friedberg §2.6, preamble to Exercises 13–17, p. 126)**.** Let V be a finite-dimensional vector space and $S \subseteq V$. The **annihilator** of S is

$$S^0 = \{\, f \in V^* : \ f(\boldsymbol{x}) = 0 \text{ for all } \boldsymbol{x} \in S \,\}.$$

When W is a subspace of V, the set W^0 is called the annihilator of W.

Proposition 5.1 (Basic properties of the annihilator; Friedberg §2.6, Exercises 13–15, p. 126–127; Homework 6, Problems 5 and 6)**.** *Let V be an n-dimensional vector space, let W, W_1, W_2 be subspaces of V, let $S \subseteq V$, and let $T : V \to U$ be linear with $\dim U < \infty$. Then*

(a) S^0 is a subspace of V^*;
(b) $\dim W + \dim W^0 = n$;
(c) if $x \notin W$, there is $f \in W^0$ with $f(x) \neq 0$;
(d) $W_1 \subseteq W_2$ implies $W_2^0 \subseteq W_1^0$; moreover $W_1 = W_2$ if and only if $W_1^0 = W_2^0$;
(e) $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$;
(f) $\mathcal{N}(T^\top) = \mathcal{R}(T)^0$.

Proof. Take a basis $x_1, \dots, x_k$ of W, extend it to a basis $\beta = \{x_1, \dots, x_n\}$ of V, and let $\beta^* = \{f_1, \dots, f_n\}$ be the dual basis. Since $f_i(x_j) = \delta_{ij}$,

$$W^0 = \operatorname{span}\{f_{k+1}, \dots, f_n\}.$$

For this identity: when $j > k$, the functional f_j vanishes on every x_i with $i \leq k$, so $f_j \in W^0$; conversely, write $f \in W^0$ as $f = \sum_{i=1}^n c_i f_i$, so that $c_i = f(x_i)$, and $c_i = 0$ for $i \leq k$ because $x_i \in W$; hence $f \in \operatorname{span}\{f_{k+1}, \dots, f_n\}$.

(a) The zero functional lies in S^0; if $f, g \in S^0$ and $c \in \mathbb{F}$, then $(f+g)(x) = 0$ and $(cf)(x) = 0$ for every $x \in S$, so S^0 is closed under addition and scalar multiplication.

(b) The identity above gives $\dim W^0 = n - k = n - \dim W$.

(c) If $x \notin W$, then x is not a combination of $x_1, \dots, x_k$, so some f_j with $j > k$ has $f_j(x) \neq 0$, and this f_j lies in W^0.

(d) If $W_1 \subseteq W_2$ and $f \in W_2^0$, then f vanishes on W_2, hence on W_1, so $f \in W_1^0$. If $W_1^0 = W_2^0$ but $W_1 \neq W_2$, say $W_1 \not\subseteq W_2$, take $x \in W_1 \setminus W_2$; by (c) there is $f \in W_2^0$ with $f(x) \neq 0$, while $x \in W_1$ forces $f \notin W_1^0$, a contradiction.

(e) A functional f vanishes on $W_1 + W_2$ exactly when it vanishes on both W_1 and W_2: it takes the value $f(x) + f(y)$ at each $x + y$, and W_1, W_2 both lie in $W_1 + W_2$.

(f) For $g \in U^*$, $T^\top(g) = 0$ exactly when $g(Tx) = 0$ for all $x \in V$, that is, when g vanishes on $\mathcal{R}(T)$, which is $g \in \mathcal{R}(T)^0$. □

Remark 5.2 (The annihilator in the row-vector picture)**.** Section 4 of Lecture 5 writes a functional as a row: once a basis β is fixed, $f(\boldsymbol{x}) = r[\boldsymbol{x}]_\beta$, and the dual basis is given by the rows of P^{-1}. The annihilator is then the set of functionals that pair to 0 with every vector of W. Extend a basis $x_1, \dots, x_k$ of W to a basis β, write its dual basis as $\beta^* = \{f_1, \dots, f_n\}$, and let $r_1, \dots, r_n$ be their coordinate rows. Then

$$W^0 = \operatorname{span}\{f_{k+1}, \dots, f_n\},$$

the functionals that read only the $n - k$ coordinates outside W; they are linearly independent, hence $\dim W^0 = n - k$. A change of basis multiplies this family of rows on the right by Q: the coordinates change, while the functionals and W^0 itself do not, and since right multiplication by an invertible matrix preserves rank, the dimension is unchanged. The kernel of the transpose can be read the same way: $T^\top(g) = g \circ T$ sends the row r to rA, so $g \in \mathcal{N}(T^\top)$ means $rA = \mathbf{0}$, that is, r pairs to 0 with every column of A, i.e. g vanishes on $\mathcal{R}(T)$, hence $g \in \mathcal{R}(T)^0$.

Definition 5.2 (The annihilator inside $\mathcal{L}(V, W)$; Homework 4, Problem 8)**.** Let V, W be vector spaces and let $S \subseteq V$. The set

$$S^0 = \{\, T \in \mathcal{L}(V, W) : \; T(\boldsymbol{x}) = \mathbf{0} \text{ for all } \boldsymbol{x} \in S \,\}$$

is also called the **annihilator** of S: the same construction as Definition 5.1, with V^* replaced by $\mathcal{L}(V, W)$.

6 Checklist and Exercises

By the end of this lecture you should be able to:

1. define a partitioned matrix, a submatrix and compatible partitions, and write $(AB)_{ij} = \sum_l A_{il}B_{lj}$ blockwise;
2. write AB as a sum of n outer products by the column–row expansion, and explain why its (i,j)-entry is $\sum_k a_{ik}b_{kj}$;
3. state the criterion for a block triangular matrix to be invertible, with the formula for the inverse,
4. compute the inverse of a 3×3 matrix by $(A\,|\,I) \to (I\,|\,A^{-1})$, and say what must appear in the left half when it fails;
5. distinguish homogeneous from nonhomogeneous systems, and state that the homogeneous solution set is $\mathcal{N}(L_A)$ with dimension $n - \operatorname{rank}(A)$;
6. write the solution set of a nonhomogeneous system as "a particular solution plus the homogeneous solution set", and say when it consists of a single point;
7. reduce an augmented matrix to reduced echelon form, read off the basic and free variables, write the particular solution and a basis of K_H, and explain why those $n - r$ vectors are linearly independent;
8. write $W^0 = \operatorname{span}\{r_{k+1}, \dots, r_n\}$ in the dual basis, deduce $\dim W^0 = n - \dim W$ from it, and say why $\mathcal{N}(T^\top) = \mathcal{R}(T)^0$ is exactly $rA = \mathbf{0}$.

Exercises (printed pages). Lay: §2.4 p. 121–122 (Exercises 1–16: block arithmetic, block diagonal matrices and Schur complements), §2.2 p. 109 (Exercises 1–4 and 29–34: computing an inverse), §2.3 p. 115 (Exercises 1–10: testing invertibility), §1.5 p. 47 (Exercises 1–16: solution sets), §1.2 p. 21 (Exercises 1–14: echelon forms); Friedberg: §3.2 p. 165 (Exercises 1–15: rank, inverses and augmented matrices), §3.3 p. 179 (Exercises 1–11: homogeneous solution sets and particular solutions), §3.4 p. 194 (Exercises 1–10: reduced echelon form and general solutions). Each section is preceded by Practice Problems; do those first.

How to use the two texts together:

- **Computation and proof divide cleanly.** How a block product is computed and how the row reduction for an inverse runs: Lay (§2.4, §2.2). Why $(A\,|\,I)$ is legitimate and why row operations preserve a solution set: Friedberg (§3.2, §3.4).
- The spine of the lecture: *blocks → block multiplication and the column–row expansion → inverses of block triangular matrices → the inverse by $(A\,|\,I) \to$ → the shape of a solution set → reading it off the reduced echelon form.*
- The central point: inverting by $[\,A \mid I\,] \to [\,I \mid A^{-1}\,]$ and solving a system are two readings of one row reduction, one reading the right half as I and the other as the right-hand side; the block matrix section offers the cheaper variant of the same work, cutting the matrix so that multiplication and inversion happen on smaller matrices.